EXCRETORY PRODUCTS AND THEIR ELIMINATION

1 EXCRETORY WASTE

METABOLISM — EXCESS INGESTION

Results in accumulation of

- **Nitrogenous wastes** – NH_3, Urea, Uric acid
- Other contents CO_2, H_2O, ions (Na^+, K^+ Cl^-, PO_4^{3-}, SO_4^{2-})

↓

Removed Partially/Completely

2 NITROGENOUS WASTES

- Nature of nitrogenous waste formed and their excertion vary among anilmals depending on tne **habitat/availability of water**

Major Nitrogenous Waste	Nature & Examples	Toxicity And Water Required	Typical
• Ammonia	**Ammonotelic**	Maximum	• Diffusion through gills surface or body surface ammonium ions
• Urea	**Ureotelic**	Less	• **Kidneys** filter urea from blood
• Uric acid	**Urecotelic** • Land snads • Insects • Reptiles • Birds	Least	Pellet/Paste

- Ammonia converts into urea in liver via ornithine cycle.
- Elimination of urea, uric acid is meant for conservation or water i.e., a type of terrestrial adaptation.
- Kidneys do not play a significant role in removal of ammonia
- Some amount of urea may be retained in the kidney
- matrix of some animals to maintain desired osmolarity

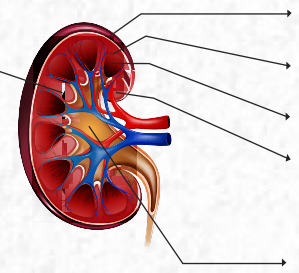
3 EXCRETORY STRUCTURES

- Most invertebrates - Simple tubular forms
- Vertebrates - Complex tubular organs called kidneys

Structures	Examples
• Protonephridia/flame cells	• Platyhelminthes (Planaria)
• Nephridia	• Rotifers
• Malpighian tubules	• Some annelids
• Antennal/Green glands	• Cephalochordates (Amphioxus)
	• Annelids (Earthworms)
	• Insects (Cockroaches)
	• Crustaceans (Prawn)

- **Protonephridia** are primarily concerned with **osmoregulation**
- **Functions of excretory structures**
- Eliminate nitrogenous wastes
- Maintan ionic and acid-base balance of body fluids. i.e., osmoregulation

5 KIDNEY



Columns of Bertini
Renal columns

- Part of cortex which extents between medullary pyramids

Capsule-outer tough covering

Cortex (Outer)

Medulla (Inner)

zones

Hilum
- Notch towards concave-surface
- Ureter, blood vessels and nerves enter

Renal pelvis
- Funnel shaped space with projection called calyes

section of kidney

Medullary pyramids are conical masses that project into calyces.

MICTURITION

- Process of release of urine
- Mechanism - **Micturition reflex**

↓

Urinary bladder (Store Urine)
Activates

↓

Stretch receptors

signals

CNS (voluntary signal)

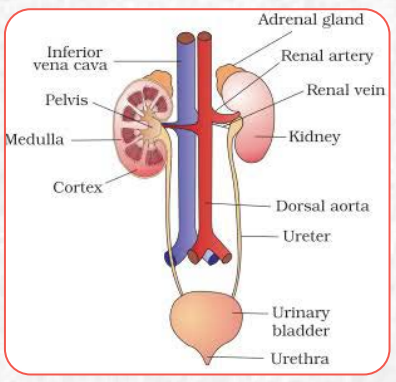
↓ Send motor message

Urinary bladder
- Smooth muscle contract
- Urethral sphincters relax

↓ Release

Urine

4 HUMAN EXCRETORY SYSTEM



KIDNEY

- 1 pair, bean shaped, reddish brown.
- Length 10-12 cm, Width 5-7 cm Thickness 2-3 cm.
- Weight 120-470 g.
- Between T_{12} – L_3 vertebra, close lo dorsal inner wall of abdominal cavity.

URINARY BLADDER

- Have stretch receptors.
- Store urine till voluntary signals from CNS carries out its release.

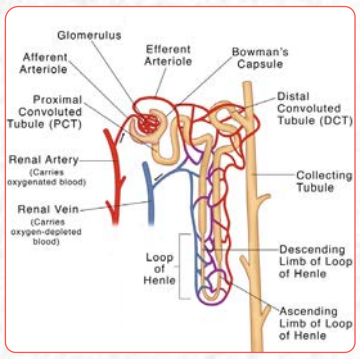
URETHRA

- Guarded by sphincters.
- Meant for release of urine.

6 NEPHRON

- **Functional unit of kidney**
- Each nephron has two parts
- Nearly **1 million/kidney**

Parts: (i) Glomerulus (ii) Renal tubule



Parameters	Cortical	Juxtamedullary
• Number	More	Less
• Loop of Henle	Too short	Very long
• Extension Into medulla	Very Mild	Deep
• Vasa recta	Absent/reduced	Present

• **Juxtaglomerular apparatus (JGA)**
Sensitive region formed by cellular modifications in **distal convoluted tubule** and **afferent arteriole** at the location of their **contact**

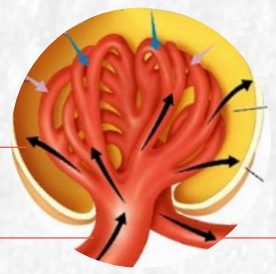
• Nephrons are dipped in interstitial fluid having specific osmolarity
- Cortex - 300 mOsm/L - Medulla - upto 1200 mOsm/L (Gradient)

• **Glomerulus** is a tuft of capillaries formed by Afferent Arteriole. A fine branch of Renal Artery.
• **Malpighian Corpuscle, PCT, DCT – Located In Cortex.**
• Loop of Henle – Dips into Medulla.
•Many Ducts open into straight tube called collecting duct, many of which converge into renal pelvis through mediullary Pyramids in the calyces.
• **Efferent Arteriole** emerging from glomerulus forms peritubular capillaries around renal tubule
• **Vesa recta:**
Branch of petitubular capillaries
Parallel to loop of Henle
'U' shaped

7 URINE FORMATION

Main Processes

1. Glomerular filtration
   180 L filtrate/day
2. Tubular reabsorption
   Near **99%** of filtrate get **reabsorbed** by renal tubules By
   - **Active process**
     • Glucose, Na^+ amino acid
   - **Passive process**
     • Nitrogenous wastes, H_2O
3. Tubular secretion
   •H^+, K^+ and ammonia secreted into filtrate
   • Maintains ionic and acid base balance



Urine 1-1.5 L/day

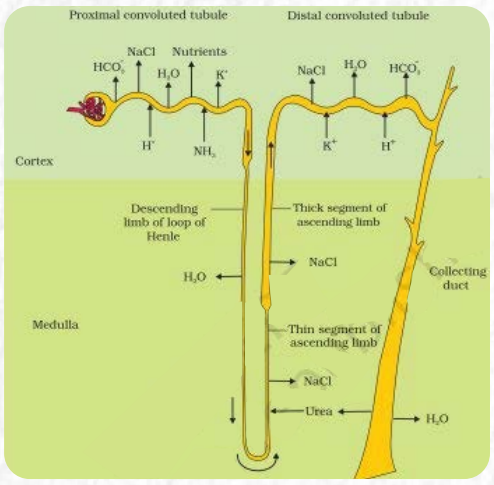
Glomerular Filtration/Ultra filtration
-1/5 of cardiac output or 1100-1200 ml blood/min is filtered by kidneys
↓
Renal Artery
↓
Arterioles
↓
Endothelium of glomerular blood vessels
↓
Basement membrane
↓
Epithelium of Bowman's capsule
↓
Glomenular filtrate (Plasma except proteins)

Filtration membrane

Bowman's capsule have **poducytes** arranged in intricate manner so as to leave some spaces called slit pores/**filtration slits**

• Filtration is due to pressure in the glomerular capillaries.
• **Glomerular filtration rate (GFR) = Filtration/min 125ml/min**
• Kidney has an ability to regulate GFR.

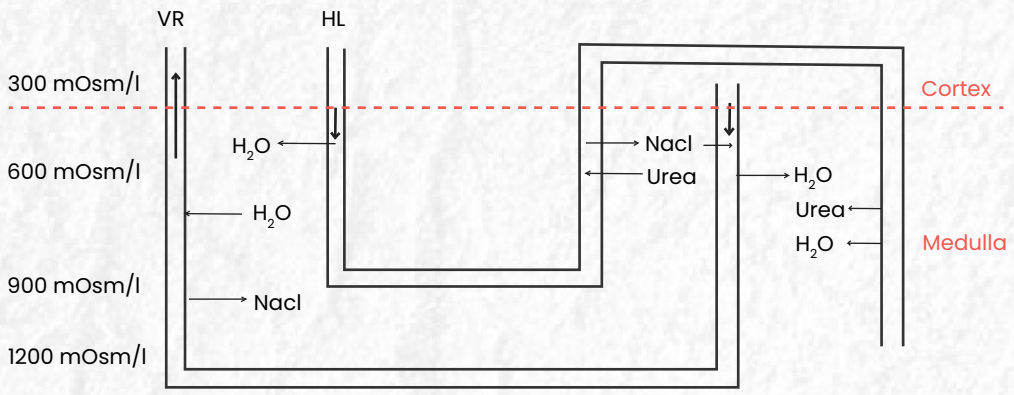
• Lined by simple cuboidal brush border epithelium
• Nearly all essential nutrients, 70-80% electrolytes and water ere reabsorbed
• Major site of reabsorption & selective secretion



8 COUNTER CURRENT MECHANISM TO CONCEITTRATE FILTRATE

- Flow of filtrate in different limbs of following structures are in opposite direction (**Counter** current)

- Loop of Henle (HL)
- Vasa Recta (VR)
- HL and VR



- Proximity of HL and VR and counter current in them increase osmolarity towards inner medullary interstitium (300 mOsm/L in cortex to 1200 mOsm/L)
- **Interstitium gradient is caused by NaCl and urea.**
- NaCl transported by ascending limb of HL exchanged with descending limb of vasa recta and is returned to medullary interstitium by ascending limb of VR.
- Urea which enters in thin part of ascending limb of HL is transported back to interstitium by collecting tubule
- This mechanism maintain interstitial concentration gradient that helps in easy passage of water from collecting tubule thereby concentrating filtrate (urine).

- Mammals have ability to poroduce concentrated urine.
- HL primarily helps to maintain osmolarity gradient in kidney intetstitium.

9 REGULATION OF KIDNEY FUNCTION/GLOMERULAR FILTRATION RATE

Hypothalamus

Low blood volume
Body fluid volume
Ionic concentration

↓

Activate

Osmoreceptors of Hypothalamus to release ADH/Vasopressin from Neurohypophysis

↓

Work

- Constrict blood vessels
- Increase reabsorption of water from DCT (Prevent diuresis)

↓

Result

- Blood volume increases
- Blood pressure increases
- GFR increases

→ **Increase in blood volume**
↓ Switch off
Osmoreceptors
↓ Suppress
ADH/Vasopressin

JGA

Low GFR/Glomerular blood flow
Glomerular blood pressure

↓

Activate

JG cells to release **renin**

↓ Liver

↓ Angiotensinogen

↓ **Angiotensin I**

↓ **Angiotensin II**

↓

work

- Constrict blood vessel
- Activate adrenal cortex to release Aldosterone, that causes reabsorption of Na^+ and water

↓

Result

- Blood pressure increases
- GFR increases

(**Renin-Angiotensin** mechanism)

Heart

Increase blood flow to atria of heart

↓

Release of ANF (Atrial natriuretic factor)

↓

work

Vasodilation

↓

Result

- Blood pressure increases decrease
- GFR increases

ANF mechanism acts as a check on Renin-Angiotensin mechanism

10 CHARACTERISTICS AND COMPOSITION OF URINE

Colour - Light yellow

pH = 6

Odour - Characteristic

Human kidneys can produce urine nearly **4 times** concentrated than initial filtrate.

Urea - 25-30 gm/day

Various conditions can affect characteristics of urine.

Abnormal constituents of urine	Condition	Indicate
Glucose	Glucosuria	Diabetes mellitus
Ketone bodies	Ketonuria	Diabetes mellitus

11 DISORDERS OF EXCRETORY SYSTEM

Disorders	Symptoms or Treatment
Renal catculi	Stone or insoluble mass of crystalised salts (e.g. oxalates)
Glomerulonephritls	Inflammation of glomeruli of kidney
Renal/kidney failure	Malfunctioning of kidneys leads to accumulation of urea in blood (Uremia); highly harmful, may lead to kidney failure. **Treatment** **(i) Haemodialysis: Process to remove urea from blood** Method is a boon for thousands of uremic patients all over the world. Blood drained from artery —Mix with→ **Heparin (Anticoagulant)** —Pumped through→ **Dialysing unit** **Artificial kidney** Porous cellophane tubes surrounded by dialysing fluid. —Filtration based on concentration gradient, Nitrogenous wastes freely move out→ Clear blood —Mix with→ Anticoagulant —Pumped back to body through Vein→ Blood **(ii) Kidney transplantation** Ultimate method in correction of **acute** renal failure • Functional kidney is taken from donor • To minimize rejection, **close relatives** are **preferred** as donor • Modern clinical problems have increased success rate of such complicated techniques
Composition of dialysis fluid is same as plasma except the nitrogenous wastes	

12 ROLE OF OTHER ORGANS IN EXCRETION

Accessory structure	substances	Basic Work
Lungs	CO_2, water	• Remove large amount of CO_2, approximately 200 mL/min • Remove significant quantity of water
Liver (Largest gland)	Bilirubin, vitamins biliverdin, drugs cholesterol, degraded steroid hormones	• Most of these substances pass out alongwith digestive wastes
Skin • Sweat glands	Sweat contains • NaCl • Urea • Lactic acid	• Primary function of sweat is to facilitate cooling effect on body surface
• Sebaceous glands	Sebum contains • Sterols • Hydrocarbons • Waxes	• Sebum provides a protective oily covering for the skin
Salivary glands		• Small amount of nitrogenous wastes are eliminated through saliva